

Assignment

MAT102, Section:3

Group-1:

1. Solve the linear differential equation for the current $i(t)$

$$L \frac{di}{dt} + Ri = E(t), \quad i(0) = i_0$$

2. Show that the given differential equation is homogeneous. Also solve the differential equation using a proper substitution.

$$x \frac{dy}{dx} = y + \sqrt{x^2 - y^2}, \quad x > 0$$

Group-2:

1. Consider the wave equation, $u_{tt} - u_{xx} = 0$ for $0 \leq x \leq 1$ and $t > 0$

Boundary Conditions: $u(0, t) = u(1, t) = 0$ for $t > 0$

Initial Conditions: $u(x, 0) = \sin 2\pi x$ and $u_t(x, 0) = 0$ for $0 \leq x \leq 1$

Find out $u(x, t)$ for all x and t

2. Solve the partial differential equation, $xp + yq = 4z$

Group-3:

1. Solve the nonlinear differential equation of the population $P(t)$

$$\frac{dP}{dt} = P(a - bP), \quad P(0) = P_0$$

2. Show that the given differential equation is Bernoulli. Also solve the differential equation using a proper substitution.

$$t^2 \frac{dy}{dt} + y^2 = ty$$

Group-4:

1. Find an explicit solution to the given initial value problem using method of separation of variables. $y' = y^2 \sin x^2$, $y(-2) = \frac{1}{3}$
2. Solve the given differential equation by using an appropriate substitution.

$$\frac{dy}{dx} = 2 + \sqrt{y - 2x + 3}$$

Group-5:

1. Consider the wave equation, $u_{tt} - u_{xx} = 0$ for $0 \leq x \leq 2$ and $t > 0$

Boundary Conditions: $u(0, t) = u(2, t) = 0$ for $t > 0$

Initial Conditions: $u(x, 0) = \sin 3\pi x + \sin 12\pi x$ and $u_t(x, 0) = 0$ for $0 \leq x \leq 2$

Find out $u(x, t)$ for all x and t

2. Solve the partial differential equation, $x^2p + y^2q - 2z^2 = 0$

Group-6:

1. Consider the wave equation, $u_{tt} - u_{xx} = 0$ for $0 \leq x \leq 3$ and $t > 0$

Boundary Conditions: $u(0, t) = u(3, t) = 0$ for $t > 0$

Initial Conditions: $u(x, 0) = (1 - 2x)$ and $u_t(x, 0) = 1$ for $0 \leq x \leq 3$

Find out $u(x, t)$ for all x and t

2. Solve the partial differential equation, $yp - xq = 2x - 3y$

Group-7:

1. Eliminate the arbitrary function and hence obtain the partial differential equation,

$$\phi(x - 2y + 2z, x^2 + y^2 + z^2)$$

2. Solve the given differential equation by undetermined coefficients.

$$y'' - 2y' + 5y = e^x \cos 2x$$

Group-8:

1. Solve the given differential equation by variation of parameters.

$$x^2y'' + xy' - y = \ln x$$

2. The given differential equation is based on the motion of a falling object from a specific height. Solve the differential equation for the height $s(t)$.

$$\frac{d^2s}{dt^2} = -g, \quad s(0) = s_0, \quad s'(0) = v_0$$

Group-9:

1. Solve the partial differential equation, $x(x^2 + 3y^2)p - y(3x^2 + y^2)q = 2z(x^2 - y^2)$

2. Verify that the given differential equation is not exact. First convert it to an exact one and then solve it. $(-xy \sin x + 2y \cos x)dx + 2x \cos x dy = 0$